Knowledge and RAG

Petra Galuscakova
petra.galuscakova@uis.no

University of Stavanger

Information Access & Artificial Intelligence

Department of Electrical Engineering and Computer Science
University of Stavanger

March 12, 2026

Overview

Outline

Limitations of Current LMs: Hallucinations

Explain Akari Asai's research, including her representative papers

(ChatGPT4o, February 8, 2025)

Akari Asai is a researcher in natural language processing (NLP) and artificial intelligence (AI), particularly focusing on open-domain question answering, knowledge retriev Her work primarily aims to improve how machines retrieve, understand, and r information efficiently.

Learning to Retrieve Passages without Supervision

Ori Ram Gal Shachaf Omer Levy Jonathan Berant Amir Globerson

Blavatnik School of Computer Science, Tel Aviv University
ori.ram@cs.tau.ac.il

Representative Papers:

NAACL 2022

1. **"Learning to Retrieve Passages without Supervision" ~~(NeurIPS 2022)~~**
   - Introduces a method for training retrieval models without labeled data, improving open-domain QA efficiency.
   - Leverages contrastive learning to align questions and relevant passages.
2. **~~"One Question, Many Answers: A Retrieval-based Multimodal QA Dataset" (EMNLP 2022)~~**

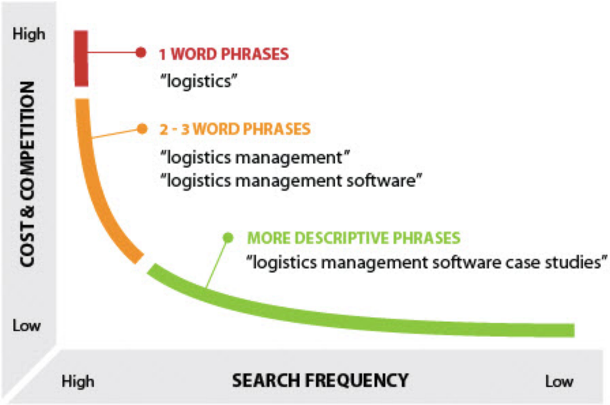
Not my paper

Venue is wrong

LMs struggle in long-tail knowledge [1]

[1] https://cmu-l3.github.io/anlp-fall2025/static_files/akari_anlp_2025_rag_lecture.pdf

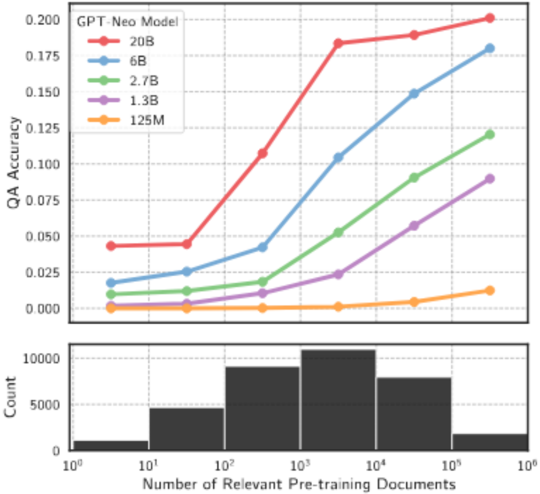
Long Tail Search Frequencies



[2]

[2] https://www.higher-education-marketing.com/blog/long-tail-keyword-strategy-ppc-seo-promote-colleges-niche-programs

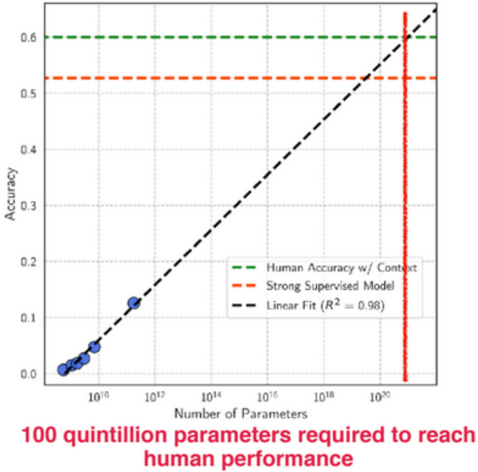
LLMs Struggle with Long Tail Knowledge



[3]

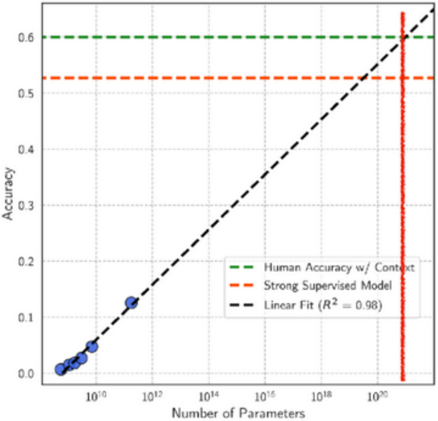
[3] https://arxiv.org/abs/2211.08411

Long-tail QA performance



[4]

[4] https://arxiv.org/abs/2211.08411



100 quintillion parameters required to reach human performance [5]

[5] https://arxiv.org/abs/2211.08411

The Knowledge Bottleneck in Language Models

- Imagine you deploy an LLM for a company assistant.
- It must answer questions about:
  - internal documents
  - technical manuals
  - recent research papers
  - company policies updated every week
- Consider the following constraints:
  - Training/retrianing the model is expensive
  - The model's knowledge becomes quickly outdated
  - You cannot verify where answers come from

Question

- How should a language model access knowledge that changes constantly?

Question

- **How should a language model access knowledge that changes constantly?**
- Possible approaches:
  - Put everything into the model weights
  - Continuously retrain the model
  - Let the model query **external knowledge sources**
- Language models do not store all knowledge, they act as reasoning engines on top of external knowledge systems.
- Retrieval-Augmented Generation (RAG)
- A hybrid between:
  - Information Retrieval systems
  - Large Language Models

Parametric vs Non-parametric memory

- Parametric memory: knowledge in weights
- Non-parametric memory: external datastore
- The combination enables scalable knowledge access

How to use external knowledge?

- Output interpolations - After solving the question yourself? [kNN LMs][6]
- Intermediate fusion – modify the LM architecture to be aware of the book? [RETRO] [7]
- Input augmentation (RAG) - Before you start solving? [RAG, REALM][8]

[6] https://arxiv.org/abs/1911.00172
[7] https://arxiv.org/pdf/2112.04426
[8] https://arxiv.org/abs/2005.11401 and https://arxiv.org/abs/2002.08909

Output Interpolations

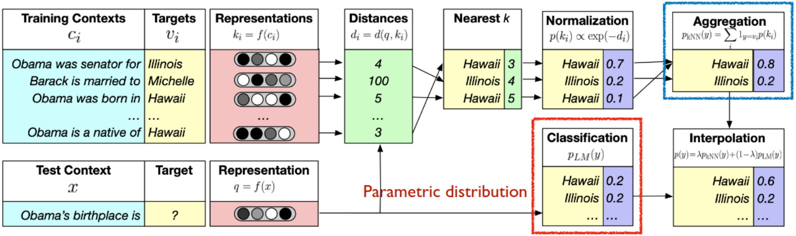
- The language model first produces predictions normally, then these predictions are adjusted using an external datastore.
- This is called output interpolation because the final probability distribution combines the LM probability and the datastore probability.
- Store hidden representations of training tokens in a vector database.
- Steps:
  - The LM processes the input normally.
  - The final hidden representation of the context is used as a query vector.
  - The system retrieves the k nearest neighbors from the datastore.
  - The tokens following those neighbors vote on the next token.
  - Combine language model probability and probability estimated from nearest neighbors

Output Interpolations (cont.)

- The LM says: Based on language statistics, the next word is probably X.
- The datastore says: In similar contexts from the training data, people used Y.

kNN-LM

Nonparametric distribution



λ: hyperparameter

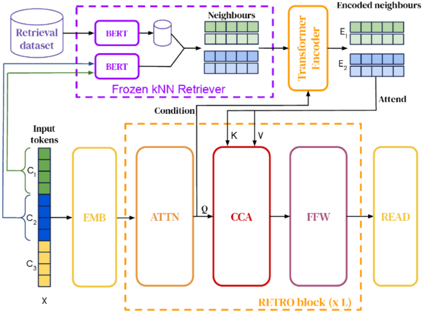
$$P_{k\mathrm{NN-LM}}(y \mid x) = (1-\lambda) P_{\mathrm{LM}}(y \mid x) + \lambda P_{k\mathrm{NN}}(y \mid x)$$ [9]

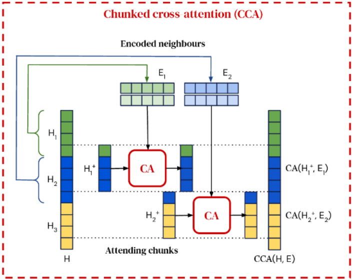
[9] https://arxiv.org/abs/1911.00172

Intermediate Fusion

- Instead of adding knowledge after prediction, the model retrieves documents during inference and processes them inside the transformer layers.
- The LM becomes retrieval-aware.
- Process:
  - Split the input into chunks.
  - For each chunk, retrieve similar chunks from a database.
  - Feed retrieved text into cross-attention layers inside the transformer.
- The model directly attends to retrieved documents while generating text.

RETRO



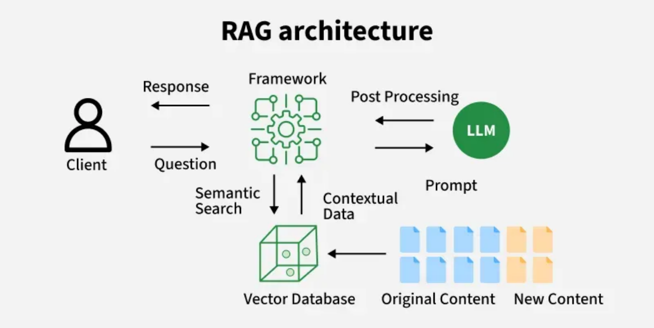


10

[10] https://arxiv.org/pdf/2112.04426

Input Augmentation

- Retrieve relevant documents before generation starts, then place them in the input context.
- The language model simply reads the retrieved information as part of the prompt.
- REALM performs retrieval during pretraining, allowing the model to learn how to use external knowledge.
- In RAG, the LM uses the provided documents to generate the answer.
- This is the most common approach today.



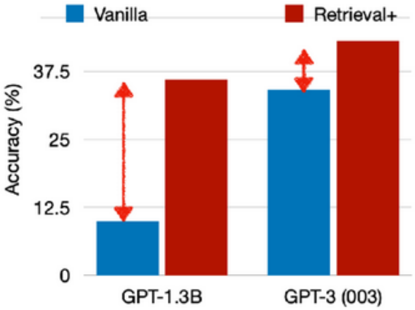
[11]

[11] https://www.geeksforgeeks.org/nlp/rag-architecture/

How retrieval-augmented LMs solve the issues?

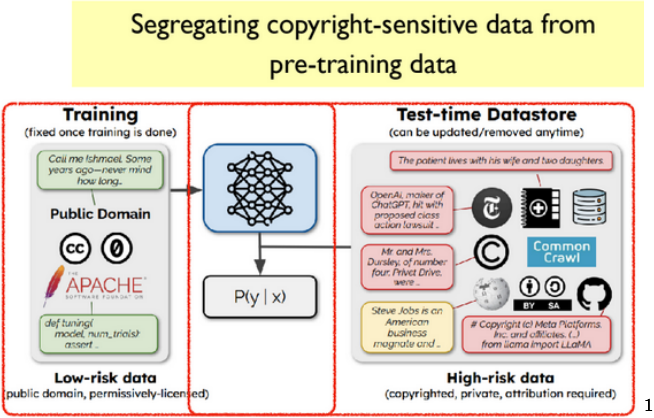
Significant improvements across model scale, with larger gain with smaller LMs

QA



[12]

[12] https://arxiv.org/abs/2212.10511



[13]

[13] https://arxiv.org/pdf/2308.04430

Outline

Retrieval Pipeline

- Need to first search and retrieve external knowledge.
- Goal: find a small subset of elements in a datastore that are the most similar to the query

sim: a similarity score between two pieces of text

Example

$$\mathrm{sim}(i,j) = \mathrm{tf}_{i,j} \times \log \frac{N}{\mathrm{df}_i}$$

\# of total docs

\# of docs containing *i*

\# of occurrences of *i* in *j*

Example

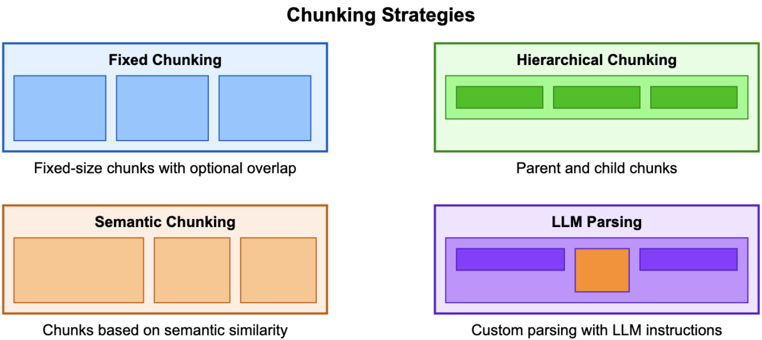
$$\mathrm{sim}(i,j) = \mathrm{Encoder}(i) \cdot \mathrm{Encoder}(j)$$

Maps the text into an *h*-dimensional vector [14]

[14] https://drive.google.com/file/d/1YUpp7L1SCK6jgdfFObsqHKXrq6HC-TLp/view

Query Formulation in RAG

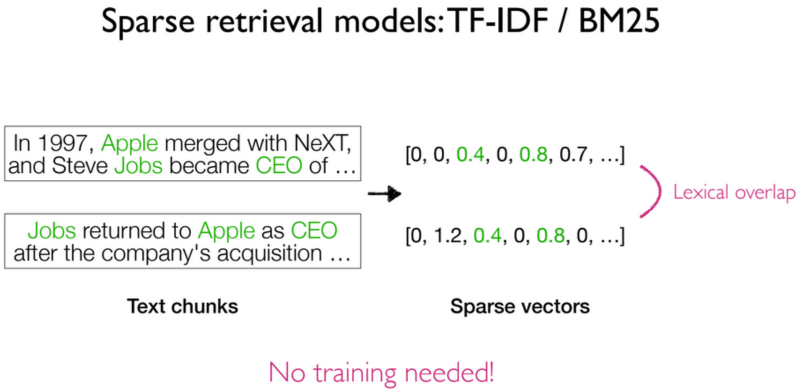
- Retrieval quality depends strongly on how the query is constructed.
- Modern RAG systems often modify the user query before retrieval.
- Common approaches:
  - Query rewriting – reformulate ambiguous queries
  - Query decomposition – split complex questions into sub-queries
  - Multi-query retrieval – generate multiple variants of the query
  - Query expansion – add related terms
  - HyDE (Hypothetical Document Embeddings) – generate a hypothetical answer and embed it
- Goal: Improve recall of relevant documents during retrieval.



[15]

[15] https://aws-samples.github.io/amazon-bedrock-samples/rag/open-source/chunking/rag_chunking_strategies_langchain_bedrock/

Sparse Retrieval



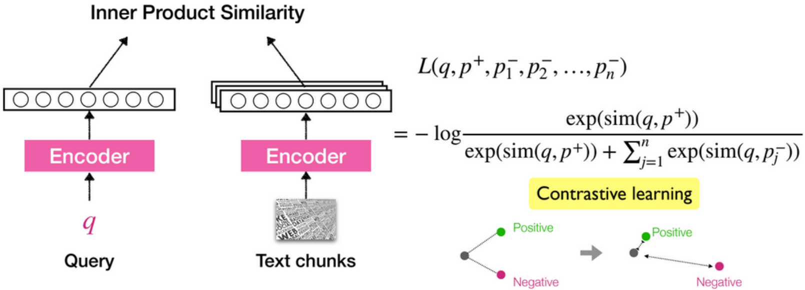
[16]

[16] Ramos: Using TF-IDF to Determine Word Relevance in Document Queries., 2003

Dense Retrieval

- Encode all documents using a LM and index using, e.g. BERT embeddings or learned embeddings
- Dense retrieval captures semantic similarity, but may miss rare terms.
- Sparse retrieval captures exact lexical matches.
- Modern systems combine both using e.g. reranking (i.e. two stage retrieval) in which fast sparse retrieval first finds possible candidates and then slower, but more precise dense retrieval sorts the results and selects the most similar documents.

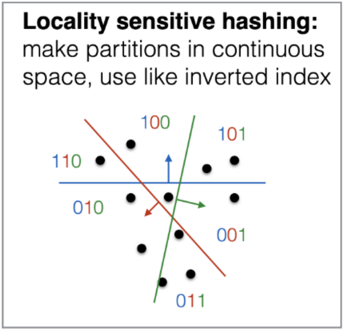
Dense Retrieval: DPR

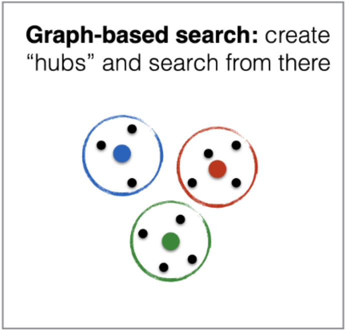


[17]

[17] https://arxiv.org/abs/2004.04906

Approximate Nearest Neighbor Search (MIPS)

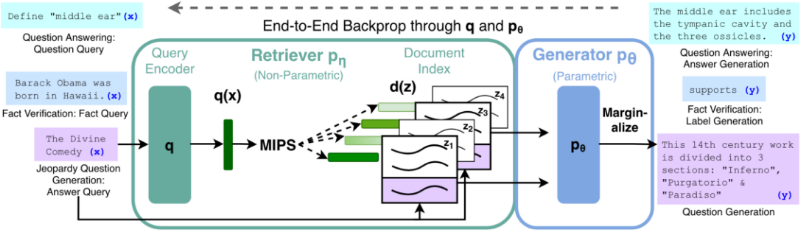




[18]

[18] https://lcs2-iitd.github.io/ELL8299-2501/static_files/ppts/21.pdf

Joint Training of Retriever and Generator



[19]

[19] https://arxiv.org/abs/2005.11401

Evaluation

- Retrieval:
  - Precision@k and Recall@k
  - MRR
  - NDCG
- Generation:
  - EM / F1
  - faithfulness
  - accuracy
- System metrics:
  - latency
  - cost
  - hallucination rate

Exercise

- Design a prompt for the LLM using the retrieved document.
- Your prompt should:
  - Use the retrieved context
  - Avoid hallucinations
  - Encourage grounded answers

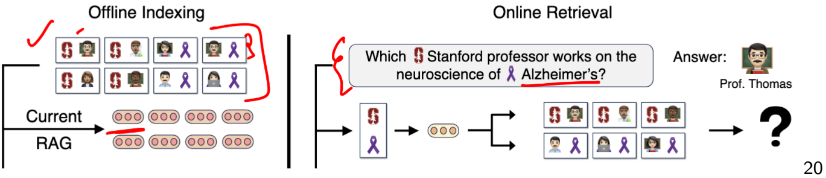
Exercise cont.

```
Based on the following context:

[CONTEXT_CHUNK_1]
[CONTEXT_CHUNK_2]
...
[CONTEXT_CHUNK_N]

Answer the following question: [USER_QUERY]
```

https://apxml.com/courses/getting-started-rag/chapter-4-rag-generation-augmentation/structuring-rag-prompts

Outline

- May fail for complex multi-hop queries.

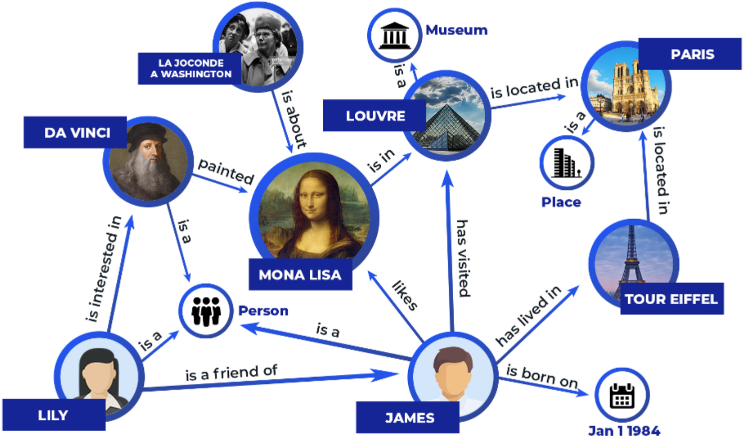


[20] http://arxiv.org/abs/2405.14831

Knowledge Graphs

- Knowledge graphs allow us to represent facts about the world in a structured form.
- Formally they are directed multi-relational graphs that represent structured data by using entities as the graph's nodes and relationships between entities as the graph's edges.

Knowledge Graphs (cont.)

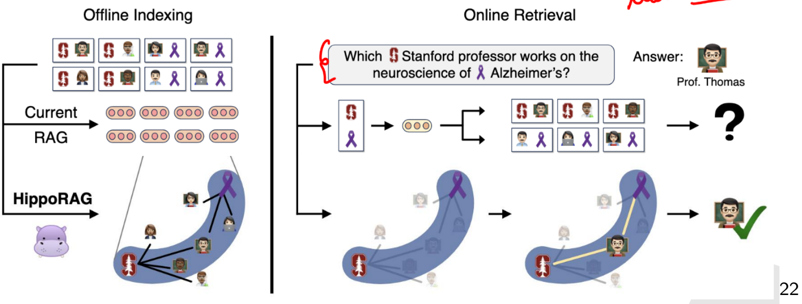


21

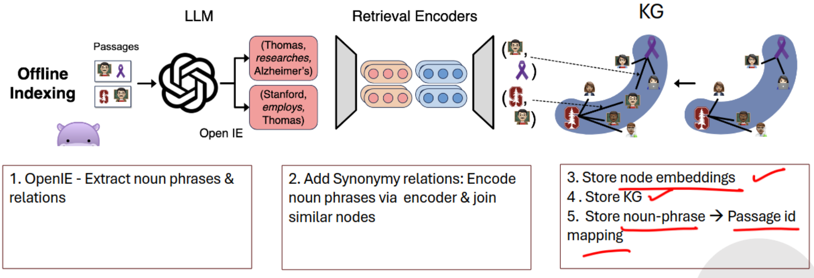
https://www.puppygraph.com/blog/knowledge-graph-examples
https://www.wikidata.org/wiki/Wikidata:Introduction

[21] https://deeppavlov.ai/research/tpost/bn15u1y4v1-improving-knowledge-graph-completion-wit

Graph RAG

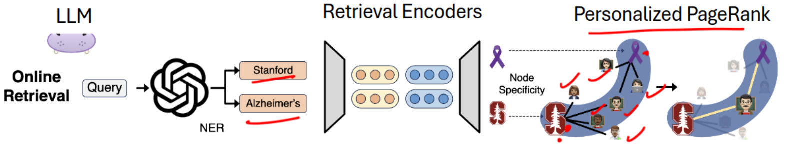


[22] https://arxiv.org/abs/2405.14831



23

[23] https://arxiv.org/abs/2405.14831



1. Extract named entities from query

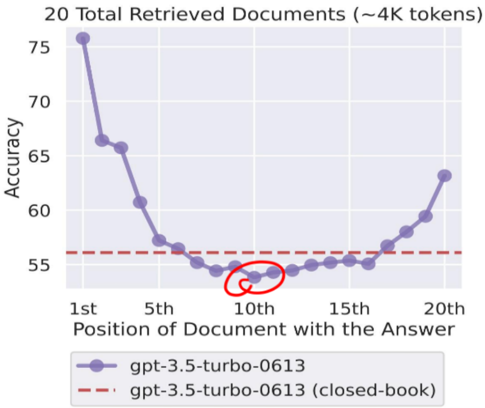
2. Encode named entities
3. Search in the index (recall we indexed node embeddings)

3. Subgraph sampling via PPR – extract nodes (noun phrases) in the nbhrhood
4. Extract the passages that mention the noun phrases [24]

[24] https://arxiv.org/abs/2405.14831

Outline

Lost in the Middle Problem

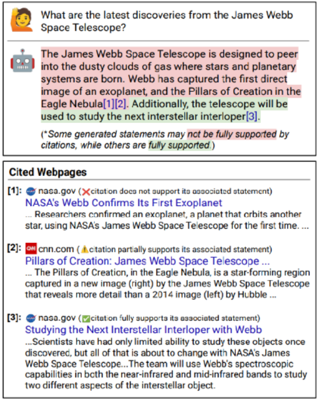


[25]

- As context increases, models miss relevant info
- Lost-in-the-middle demonstrates that models pay less attention to things in the middle of context windows

[25] https://aclanthology.org/2024.tacl-1.9/

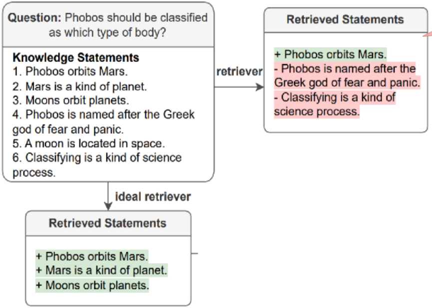
Retrieval-augmented LMs can still hallucinate



[26]

[26] https://arxiv.org/abs/2304.09848

Retrieval Failures

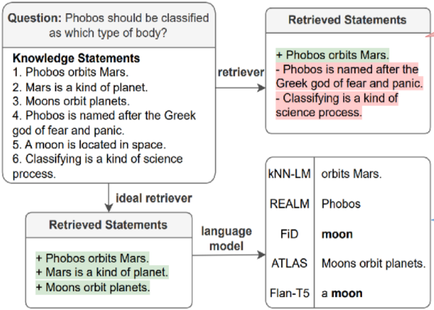


- Retrieval fails to fetch correct information.

[27]

[27] https://arxiv.org/abs/2212.09146

Reasoning Failures



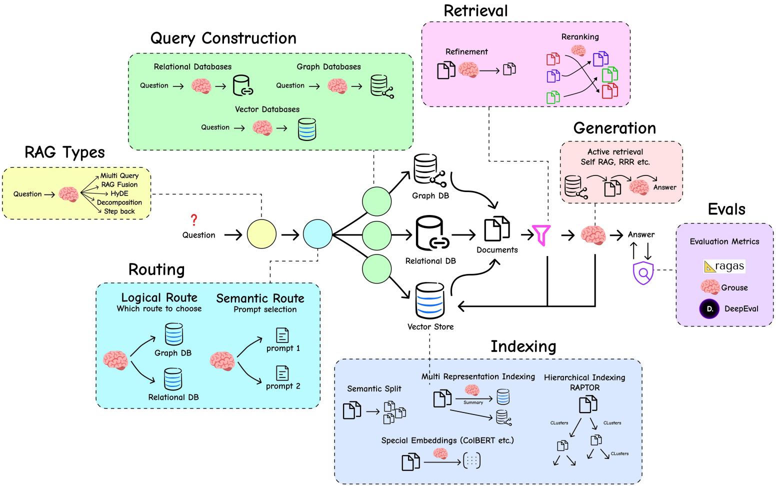
- Retrieval fails to fetch correct information.

- Even with ideal retriever, LM fails to give right answer.

[28]

[28] https://arxiv.org/abs/2212.09146

Outline



29

[29] https://levelup.gitconnected.com/building-the-entire-rag-ecosystem-and-optimizing-every-component-8f23349b96a4

Exercise

https://huggingface.co/learn/cookbook/advanced_rag

Tools

- Retrieval and Indexing (Vector databases):
  - FAISS
  - Weaviate
  - Milvus
  - Pinecone
- Classical IR systems:
  - Elasticsearch
  - Apache Lucene
- LLM / RAG Frameworks
  - LangChain
  - LlamaIndex
  - Haystack

Resources

- Akari Asai: https://drive.google.com/file/d/1YUpp7L1SCK6jgdfFObsqHKXrq6HC-TLp/view
- Yatin Nandwani: https://lcs2-iitd.github.io/ELL8299-2501/static_files/ppts/20.pdf
- Yatin Nandwani: https://lcs2-iitd.github.io/ELL8299-2501/static_files/ppts/21.pdf